

Case Study Assignment: Time Series Analysis of Disease Outbreak

Objective: Analyze a time series dataset of an infectious disease outbreak and:

- Identify trend, seasonality, and noise
- Apply forecasting models
- Compare model performance

Case Background

A health department is monitoring a fictional outbreak called “X-Flu”.
Daily reported cases show:

- Increasing trend (spread of disease)
- Seasonality (reporting delays/patterns)
- Random fluctuations

You are given **84 days of data**.

Assignment Tasks

1: Visualization

Plot Day vs Cases; Identify: Trend, Seasonality and/or Noise

Write **4–5 lines explaining patterns**

2: Decomposition

Apply **STL decomposition** or moving average

Extract: Trend, Seasonal, Residual

Explain what each component represents

3: Model Implementation

Apply **any two models**:

- Moving Average
- Exponential Smoothing (ETS)
- Polynomial Regression (trend + seasonality)
- ARIMA / SARIMA

For each:

- Fit model
- Plot **Actual vs Predicted**

4: Forecasting

Split data: Train → first 70 days, Test → last 14 days, Forecast next 14 days

5: Model Evaluation

Compute:

- MAE
- RMSE

Compare:

- Accuracy
- Ability to capture trend/seasonality

6: Residual Diagnostics

Plot residuals, Plot ACF of residuals

Comment: Are residuals white noise?

7: Final Conclusion

Answer:

- Which model performed best?
- Why?
- Limitations of your approach

Expected Tools

- Python
- Libraries:
 - pandas
 - matplotlib
 - seaborn
 - statsmodels

Submission Format

PDF file containing the code, its output, and the corresponding interpretation is provided below.